CH3SVC_VT_FuSa_ECU_RS

Functional Safety Requirements for HOST SYSTEM connected to the Camera Head

Version: 18.0
Printed by: RCT2BP
Printed on: 2026. július 2., csütörtök

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ID	Object Number	Description	Object Type	Safety Relevance	Safety Integrity	Fusa ReqType
FuSa_HS_RS_227	1	**1 Scope**	Heading	No	-	-
FuSa_HS_RS_269	1.0-1	Purpose of this document is to describe the safety requirements allocated to the HOST SYSTEM and the external interfaces of the CAMERA HEAD	Description	No	-	-
FuSa_HS_RS_373	1.0-2	This document and its content are Bosch proprietary and confidential. May not be made accessible for an unauthorized third party.	Description	No	-	-
FuSa_HS_RS_283	1.1	**1.1 Project Details**	Heading	No	-	-
FuSa_HS_RS_284	1.1.0-1	**Project Name: Camera Head 3MP SVC** (VT CH3 SVC) • Customer: AB Volvo • Application: Surround View Camera System • Area of usage: commercial vehicles • SOP: 2027	Description	No	-	-
FuSa_HS_RS_376	1.1.0-2	Functional Safety Manager: Amanda Leila AOUASSAR (XC-AC/ECT2) Functional Safety Manager: Suvin VASU (XC-AC/ECT2) (from 15th April 2026) Functional Safety Hardware Expert: Tibor RACZ (XC-AC/ENH2-Bp)	Description	No	-	-
FuSa_HS_RS_563	1.2	**1.2 Applicable Standards**	Heading	No	-	-
FuSa_HS_RS_564	1.2.0-1	ISO 26262 (2nd Edition, 2018-12) - *Road vehicles — Functional safety*	Description	No	-	-

ID	Object Number	Description	Object Type	Safety Relevance	Safety Integrity	Fusa ReqType
FuSa_HS_RS_552	2	**2 Definitions and Abbreviations**	Heading	No	-	-
FuSa_HS_RS_515	2.1	**2.1 Attribute Definitions**	Heading	No	-	-
FuSa_HS_RS_517	2.1.0-1	**Attribute Name:** Object Type **Possible values:** • "*Heading*": headers in the document structure. Not a requirement, only for documentation purposes. Implementation and verification is not required for the object (not applicable) • "*Description*": Additional information that help to understand the items, context, meaning/operation of the Implementation and verification is not required for the object (not applicable) • "*Requirement*": Requirement that shall be implemented by the HOST SYSTEM • "*Assumption* of Use": a condition that shall be fulfilled by the HOST SYSTEM	Requirement	No	-	-
FuSa_HS_RS_549	2.1.0-2	**Attribute Name:** Safety Relevance **Possible values:** • "*no*": the object is not safety related, or safety relevance is not applicable (e.g.: heading) • "*FUSA*": the object is safety relevant, it is required to fulfil the Functional Safety Requirements	Requirement	No	-	-
FuSa_HS_RS_551	2.1.0-3	**Attribute Name:** Safety Integrity Level **Possible values**: • "*QM*": the object shall be implemented in QM complaint. • "*ASIL-B*": the object shall be implemented in ASIL-B complaint • "*ASIL-B (use-case dependent)*": the necessity and the ASIL integrity of the object depends on the use-cases in the HOST SYSTEM. 1) If the HOST SYSTEM requires the given "service" with ASIL-B integrity, then the object shall be implemented in ASIL-B complaint. 2) If required but only on QM level, then implement with QM integrity. 3) If the HOST SYSTEM does not require/does not use then can be omitted.	Requirement	No	-	-
FuSa_HS_RS_450	2.2	**2.2 Definitions**	Heading	No	-	-
FuSa_HS_RS_422	2.2.0-1	OVERALL SYSTEM = The union of the HOST SYSTEM and CAMERA HEAD that implements the vehicle level function. HOST SYSTEM = The system(s) that control the CAMERA HEAD and receive its video stream. It can be a single ECU or multiple ECUs. CH = CAMERA HEAD	Description	No	-	-
FuSa_HS_RS_514	2.2.0-2	OFF time = the Camera HEAD is not powered. ON time = the Camera HEAD is electrically powered, not defined whether it streams video or not.	Description	No	-	-
FuSa_HS_RS_536	2.2.0-3	Power Cycle = a period that starts when the CAMERA HEAD leaves the unpowered state (electrical power switched ON at the Power Over Coax) and finished when the CAMERA HEAD return to de-energized state (electrical power is switched OFF at Power Over Coax)	Description	No	-	-
FuSa_HS_RS_377	2.3	**2.3 Abbreviations**	Heading	No	-	-
FuSa_HS_RS_378	2.3.0-1	CV = Computer Vision EEPROM = Electrically Erasable Programmable Read-Only Memory Embedded data = additional data inserted into the Image Frame, usually before or/and after the active image pixels. FuSa = Functional Safety GMSL = Gigabit Multimedia Serial Link (Proprietary of Analog Devices) HFPN = Horizontal Fixed Pattern Noise (related to Imager chip) HV = Human Vision ISP = Image Signal Processing MV = Machine Vision PMIC = Power Management IC PoC = Power Over Coax SoC = System on Chip	Description	No	-	-
FuSa_HS_RS_379	2.3.0-2	Abbreviations according to the ISO26262 (2nd Edition, 2018) AoU = Assumption of Use E/E = Electrical or Electronic FHTI = Fault Handling Time Interval FRTI = Fault Reaction Time Interval FTTI = Fault Tolerant Time Interval SM = Safety Mechanism TSC = Technical Safety Concept TSR = Technical Safety Requirement	Description	No	-	-

ID	Object Number	Description	Object Type	Safety Relevance	Safety Integrity	Fusa ReqType
FuSa_HS_RS_231	3	**3 Overview and Boundary of CAMERA HEAD**	Heading	No	-	-
FuSa_HS_RS_467	3.0-1	Intended usage of CAMERA HEAD: • create image about the target scene and send to HOST SYSTEM via high-speed serial link. (ASIL-B) • support HOST SYSTEM level functionality (safety relevance depends on the HOST SYSTEM expectations)	Description	No	-	-
FuSa_HS_RS_466	3.0-2	CAMERA HEAD consists of an image sensor, serializer, PMIC and an EEPROM. It is connected to the HOST SYSTEM via a single coaxial cable, that realizes both the power supply and data communication: * Image data from CAMERA HEAD to HOST SYSTEM via the co-axial cable. * Bidirectional control data (GPIO and pass through I2C messages)	Description	No	-	-
FuSa_HS_RS_468	3.1	**3.1**	Description	No	-	-
FuSa_HS_RS_490	3.1-1	CAMERA HEAD - Architectural Design of Electronics See Figure 1 on page 14.	Description	No	-	-
FuSa_HS_RS_445	3.1-2	CAMERA HEAD is designed to FAIL SAFE reaction: in case of any safety relevant failure, it stops the operation. (often called fail-stop or fail-passive) CAMERA HEAD reaches the safe state within the acceptable time interval, therefore no emergency operation mode is defined in the project. (Clause 7.4.2.6, ISO 26262-3:2018(E))	Description	No	-	-
FuSa_HS_RS_492	3.1-4	Boundaries of CAMERA HEAD for Functional Safety (ISO26262) scope (focus only the E/E parts): 1) Pin of the Coaxial Connectors: 2) The "optical path" started at the Imager Chip. note: the Functional Safety ISO26262 focuses on the malfunction of E/E systems therefore the scope of FuSa activites are reduced for the electronics.	Description	No	-	-
FuSa_HS_RS_374	3.2	**3.2 Responsibilities**	Heading	No	-	-
FuSa_HS_RS_446	3.2.0-1	Bosch is responsible for: - PMIC Configuration Settings - Imager Configuration Settings (Imager set-file) - EEPROM content Volvo is responsible for: - Serializer configuration	Description	No	-	-

ID	Object Number	Description	Object Type	Safety Relevance	Safety Integrity	Fusa ReqType
FuSa_HS_RS_233	4	**4 Requirements**	Heading	No	-	-
FuSa_HS_RS_327	4.0-1	This section describes requirements that are allocated for the HOST SYSTEM, means these requirements shall be implemented by the HOST SYSTEM.	Description	No	-	-
FuSa_HS_RS_553	4.1	**4.1 Support Functions for HOST SYSTEM**	Heading	No	-	-
FuSa_HS_RS_358	4.1.1	**4.1.1 Intrinsic Calibration Data**	Heading	No	-	-
FuSa_HS_RS_470	4.1.1.0-1	*Image Rectification* is a process during sequence of transformation are performed on the image frame to correct geometric distortions and/or align it. It can consider distortions caused by different causes like intrinsic, extrinsic etc... *Intrinsic Calibration Data* is a set of parameters used during the Image Rectification. This parameter set is unique for each piece of CAMERA HEAD.	Description	No	-	-
FuSa_HS_RS_543	4.1.1.0-2	CAMERA HEAD stores the *Intrinsic Calibration Data* and can provide to the HOST SYSTEM. The data can read out via I2C commands to CAMERA HEAD passed through via GMSL interface. CAMERA HEAD supports ASIL-B integrity for the data. CRC Checksum is available to ensure the integrity of data.	Description	No	-	-
FuSa_HS_RS_545	4.1.1.1	**4.1.1.1 Usage and Safety Relevance of Intrinsic Calibration Data**	Heading	No	-	-
FuSa_HS_RS_544	4.1.1.1.0-1	The *Intrinsic Calibration Data* is not required for the ASIL-B complaint operation of the CAMERA HEAD electronics. It is a set of parameters stored in the CAMERA HEAD but not used by the CAMERA HEAD.	Description	No	-	-
FuSa_HS_RS_539	4.1.1.1.0-2	Bosch assumes the HOST SYSTEM owner knows • The purpose of the Intrinsic Calibration Data • Reason and effects of the Calibration • How to apply the Intrinsic Calibration Data for Image Rectifying.	Assumption of Use	FUSA	ASIL B (use-case dependent)	IF
FuSa_HS_RS_540	4.1.1.1.0-3	The necessity of *Intrinsic Calibration* is application dependent (vehicle level use-cases) HOST SYSTEM owner is responsible to decide: 1) Is *Intrinsic Calibration Data* is required and in which use-cases (Human Vision, Video Perception etc..)? 2) Is it safety relevant and in which use-cases? (see Note 1)) Note 1) HOST SYTEM owner is responsible to analyse the effects of failures in the "parameter set" itself and failures in the usage of Intrinsic Calibration Data	Assumption of Use	FUSA	ASIL B (use-case dependent)	IF
FuSa_HS_RS_584	4.1.1.1.0-4	CAMERA HEAD does not have any Functional Safety expectation or requirement about Image Rectification process in the HOST SYSTEM. CAMERA HEAD does not have any responsibility about Image Rectification process in the HOST SYSTEM, except providing the *Intrinsic Calibration Data* if needed.	Assumption of Use	FUSA	ASIL B (use-case dependent)	IF
FuSa_HS_RS_567	4.1.1.2	**4.1.1.2 Applying Intrinsic Calibration in the HOST SYSTEM**	Heading	No	-	-
FuSa_HS_RS_541	4.1.1.2.0-2	HOST SYSTEM ensures the *Intrinsic Calibration Data* used for the Image rectifying is intact, correct and complete. Example for failure modes (but not the complete list): • corrupted during either temporary or permanent (non-volatile) storage in the HOST SYSTEM, • corrupted during data transfer inside the HOST SYSTEM • faulty processing by the HOST SYSTEM (e.g.: two parameters are swapped during processing).	Assumption of Use	FUSA	ASIL B (use-case dependent)	IF
FuSa_HS_RS_542	4.1.1.2.0-3	HOST SYSTEM ensures the applied *Intrinsic Calibration Data* belongs to the CAMERA HEAD that is source of received video stream. (The Intrinsic Calibration data is unique for each piece of Camera Head.) Example for failure modes (but not the complete list): • multiple CAMERA HEAD connected in the system in the same time and the calibration data are unintentionally swapped. • CAMERA HEAD is replaced in the repair shop but the HOST SYSTEM incorrectly uses the calibration data belong to old Camera Head.	Assumption of Use	FUSA	ASIL B (use-case dependent)	IF
FuSa_HS_RS_546	4.1.1.3	**4.1.1.3 Read Out the Intrinsic Calibration Data**	Heading	No	-	-
FuSa_HS_RS_575	4.1.1.3.0-1	This section is valid if the *Intrinsic Calibration Data* is read from the CAMERA HEAD and it is safety relevant. In this case the "Assumption of Use" and "Requirement" shall be fulfilled in this section.	Description	No	-	-
FuSa_HS_RS_345	4.1.1.3.0-2	HOST SYSTEM sends the correct I2C transactions to read the *Intrinsic Calibration Data* which including the use of correct memory sector address. The *Intrinsic Calibration Data* stored in the EEPROM in the following memory sector(s): Document: **[2]** (Camera Interface Specification) Section: 6.3. EEPROM MAP Block Name: Intrinsic data block (F-Theta)	Assumption of Use	FUSA	ASIL B (use-case dependent)	IF

ID	Object Number	Description	Object Type	Safety Relevance	Safety Integrity	Fusa ReqType
FuSa_HS_RS_353	4.1.1.3.0-3	HOST SYSTEM shall check the integrity of the *Intrinsic Calibration Data* (after read out but before the usage) with help of the CRC Checksum. The Checksum is stored in the EEPROM memory together with *Intrinsic Calibration Data*. If the CRC Checksums do NOT match, then HOST SYSTEM shall ensure the achievement the safe state.	Requirement	FUSA	ASIL B (use-case dependent)	SM
FuSa_HS_RS_360	4.1.2	**4.1.2 Unique Identification Number**	Heading	No	-	-
FuSa_HS_RS_441	4.1.2.0-1	*The "Unique Identification Number*" enables to uniquely identify the CAMERA HEAD if required by the HOST SYSTEM. Examples for use-cases: 1) HOST SYSTEM receives video stream from multiple CAMERA HEADs and wants to identify or verify the source of the video stream. 2) The *Intrinsic Calibration Data* stored permanently in the HOST SYSTEM instead of reading out from CAMERA HEAD at each power start-up. To ensure the connected CAMERA HEAD still matches with the stored *Intrinsic Calibration Data* , the "*Unique Identification Number"* of CAMERA HEAD can be used. (e.g. camera head replaced in a repair shop)	Description	No	-	-
FuSa_HS_RS_577	4.1.2.0-2	CAMERA HEAD stores the *"Unique Identification Number"* and can provide to the HOST SYSTEM. The data can read out via I2C commands sent to CAMERA HEAD via GMSL interface. CAMERA HEAD supports ASIL-B integrity for the data. CRC Checksum is available to ensure the integrity of data.	Description	No	-	-
FuSa_HS_RS_578	4.1.2.1	**4.1.2.1 Usage and Safety Relevance of Unique Identification Number**	Heading	No	-	-
FuSa_HS_RS_547	4.1.2.1.0-1	The "*Unique Identification Number*" is not required for the ASIL-B complaint operation of the CAMERA HEAD electronics. It is a data stored in the CAMERA HEAD, but not used by the CAMERA HEAD itself.	Description	No	-	-
FuSa_HS_RS_579	4.1.2.1.0-2	HOST SYSTEM owner is responsible to evaluate whether the "*Unique Identification Number*" is required and safety relevance in HOST SYSTEM.	Assumption of Use	FUSA	ASIL B (use-case dependent)	IF
FuSa_HS_RS_576	4.1.2.2	**4.1.2.2 Read Out the Unique Identification Number**	Heading	No	-	-
FuSa_HS_RS_346	4.1.2.2.0-1	HOST SYSTEM sends the correct I2C transactions to read out "*Unique Identification Number*" which use of correct memory sector address. The *Unique Identification Number* stored in the EEPROM in the following memory sector(s): Document: **[2]** (Camera Interface Specification) Section: 6.3. EEPROM MAP Block Name: Label data block Parameter name: Imager-ID	Assumption of Use	FUSA	ASIL B (use-case dependent)	IF
FuSa_HS_RS_354	4.1.2.2.0-2	HOST SYSTEM shall check the integrity of the "*Unique Identification Number*" (after read out but before the usage) with help of the CRC Checksum. The Checksum is stored in the EEPROM memory together with *Unique Identification Number.* If the CRC Checksums do NOT, match then HOST SYSTEM shall ensure the achievement the safe state.	Requirement	FUSA	ASIL B (use-case dependent)	SM
FuSa_HS_RS_361	4.1.3	**4.1.3 Image Synchronization**	Heading	No	-	-
FuSa_HS_RS_589	4.1.3.0-1	CAMERA HEAD can synchronize the image capturing to en external signal generated by the HOST SYSTEM (called also FRAME_SYNC). Signal: • MFP1 pin of Serializer • Name: CFG0_FSYNC	Description	No	-	-
FuSa_HS_RS_587	4.1.3.1	**4.1.3.1 Usage and Safety Relevance in HOST SYSTEM**	Heading	No	-	-
FuSa_HS_RS_588	4.1.3.1.0-1	The necessity of *Image Synchronization* is application dependent (vehicle level use-cases) HOST SYSTEM owner is responsible to decide: 1) Is *Image Synchronization* is needed and in which use-cases (Human Vision, Video Perception etc..)? 2) Is it safety relevant and in which use-cases? (see Note 1)) Note 1) HOST SYTEM owner is responsible to analyze the safety consequences of failures incorrect or missing FRAME_SYNC signal.	Assumption of Use	FUSA	ASIL B	IF
FuSa_HS_RS_590	4.1.3.2	**4.1.3.2 Requirement for Image Synchronization**	Heading	No	-	-

ID	Object Number	Description	Object Type	Safety Relevance	Safety Integrity	Fusa ReqType
FuSa_HS_RS_503	4.1.3.2-1	WHEN: continuously during video stream if it used for safety related purposes	Requirement	FUSA	ASIL B (use-case dependent)	IF
FuSa_HS_RS_351	4.1.3.2-2	HOST SYSTEM control the signal "FRAME_SYNC" corrrectly (correct timing, period time, duty cycle, polarity). CAMERA HEAD can SYNC Image Capturing to external sync signal but has not responsible to check or evaluate whether the HOST SYSTEM send correct signal. If wrong F_SYNC is sent by HOST SYSTEM, CAMERA HEAD may follow it. background: incorrect behaviour of this signal in the ECU will cause incorrect timing of the Image Capturing in the Camera Head (delayed picture) For specification (timing, pulse width, levels): see Document **[2]** - "Camera Interface Specification"	Assumption of Use	FUSA	ASIL B (use-case dependent)	IF
FuSa_HS_RS_436	4.1.3.2-3	IF *Image Synchronization* is NOT USED: the FRAME_SYNC signal shall be configures permantly to the state described in the Document **[2]** - "Camera Interface Specification".	Requirement	FUSA	ASIL B (use-case dependent)	Other
FuSa_HS_RS_555	4.2	**4.2 States**	Heading	No	-	-
FuSa_HS_RS_534	4.2.0-1	To ensure ASIL-B complaint operation the CAMERA HEAD shall be started from the following defined state: • a de-energized state (power input on Power Over Coax is switched OFF). note: this inital state ensure all electronics components has defined state and conditions.	Requirement	FUSA	ASIL B	IF
FuSa_HS_RS_348	4.2.0-2	HOST SYSTEM shall NOT use the outputs of CAMERA HEAD for safety relevant purposes (video stream or EEPROM data) until: • all configurations done successfully • all Start-up tests passed successfully • all Safety Mechanisms activated successfully in the CAMERA HEAD (Imager, Serializer) • all Safety Mechanisms in HOST SYSTEM that are CAMERA HEAD relevant activated successfully (content of this document) Exception: the start-up tests and safety mechanisms belong to the Imager chip or Image Stream are not required for the process "Read out from EEPROM".	Requirement	FUSA	ASIL B	IF
FuSa_HS_RS_583	4.2.0-3	According to the requirement "FuSa_HS_RS_348" the following operating states cannot use for safety purposes, HOST SYSTEM is responsible ensure the safe state: • *Start-up phase of the CAMERA HEAD* (before the start-up tests passed and safety mechanism are activated) • *Shutdown phase of the CAMERA HEAD* (as soon any of the safety mechanism stopped either in CAMERA HEAD or in the HOST SYSTEM) • *Special operation modes* where any of the safety mechanism is not active either in the CAMERA HEAD or in HOST SYSTEM (if applicable) (e.g.: test mode)	Requirement	FUSA	ASIL B	IF
FuSa_HS_RS_489	4.2.1	**4.2.1 Recovering from Safe States**	Heading	No	-	-
FuSa_HS_RS_447	4.2.1.0-1	To recover CAMERA HEAD from safe state the HOST SYSTEM shall switch OFF the power supply of Power Over Coax. The CAMERA HEAD operation can start when the "REQ ID: FuSa_HS_RS_534" is fulfilled.	Requirement	No	-	-
FuSa_HS_RS_362	4.3	**4.3 General Requirements**	Heading	No	-	-
FuSa_HS_RS_556	4.3.1	**4.3.1 EEPROM**	Heading	No	-	-
FuSa_HS_RS_334	4.3.1.0-2	HOST SYSTEM shall not modify, overwrite or delete any data in the CAMERA HEAD EEPROM This requirement is NOT valid for the memory block that is dedicated for the HOST SYSTEM usage: Document: [2] (Camera Interface Specification) Section: 6.3. EEPROM MAP Block Name: (No. 6) OEM Data Block	Assumption of Use	FUSA	ASIL B	IF
FuSa_HS_RS_524	4.3.1.0-3	There is a memory block that dedicated for the HOST SYSTEM usage and arbitrary data can be written by the HOST SYSTEM. Document: [2] (Camera Interface Specification) Section: 6.3. EEPROM MAP Block Name: (No. 6) OEM Data Block **WARNING:** ASIL-B capability of the CAMERA HEAD is not NOT valid for the storage of this data.	Description	No	-	-
FuSa_HS_RS_566	4.3.1.0-4	The *"Write Protect"* functionality of EEPROM (signal EEPROM_WP) is connected to the Serializer GPIO (MFP 8) and can read/control by the HOST SYSTEM. It is not used in the Functional Safety Concept of CAMERA HEAD, not safety relevant. HOST SYSTEM can decide whether use this signal or not.	Description	No	-	-

ID	Object Number	Description	Object Type	Safety Relevance	Safety Integrity	Fusa ReqType
FuSa_HS_RS_573	4.3.2	**4.3.2 PMIC**	Heading	No	-	-
FuSa_HS_RS_444	4.3.2.0-1	The "*ERROR"* signal of the PMIC (*PMIC_ERROR_N*) is connected to the Serializer GPIO (MFP5) and can read out by the HOST SYSTEM. It is not used in the Functional Safety Concept of CAMERA HEAD, not safety relevant. HOST SYSTEM can decide whether is it used for general error diagnostic.	Description	No	-	-
FuSa_HS_RS_574	4.3.3	**4.3.3 Imager**	Heading	No	-	-
FuSa_HS_RS_533	4.3.3.0-1	The ERROR signal of the Imager (*IMG_ERROR_N*) is connected to the Serializer GPIO (MFP7) and can read out by the HOST SYSTEM. It is not used in the Functional Safety Concept of CAMERA HEAD, not safety relevant. HOST SYSTEM can decide whether is it used for general error diagnostic.	Description	No	-	-
FuSa_HS_RS_557	4.3.4	**4.3.4 I2C Interface of Camera Head**	Heading	No	-	-
FuSa_HS_RS_565	4.3.4.0-1	CAMERA HEAD has an internal I2C bus that provides communication towards to the connected 4 ICs. Unique I2C addresses makes possible the individual communications with each chip. This I2C is accessible for the HOST SYSTEM with help of the "I2C Pass through" function of the Serializer.	Description	No	-	-
FuSa_HS_RS_527	4.3.4.0-2	HOST SYSTEM shall always use the correct I2C address belong to the target IC (EEPROM, PMIC, Serializer or Imager). background: wrong I2C address causes incorrect messages sent, lost or unintentionally inserted I2C messages.	Assumption of Use	FUSA	ASIL B	IF
FuSa_HS_RS_558	4.3.4.0-3	HOST SYSTEM shall use the correct I2C speed that is supported by the CAMERA HEAD	Assumption of Use	FUSA	ASIL B	IF
FuSa_HS_RS_538	4.3.4.0-4	HOST SYSTEM shall check the ACK of I2C messages. If ACK is NOT received, the HOST SYSTEM shall ensure the necessary safety measure or ensure the safe state.	Assumption of Use	FUSA	ASIL B	SM
FuSa_HS_RS_335	4.3.5	**4.3.5 Power Supply of Camera Head**	Heading	No	-	-
FuSa_HS_RS_339	4.3.5.0-1	HOST SYSTEM shall ensure the power supply provided for the Camera Head (Power Over Coax) is according to the specifications described in the following document: Document **[8]** - Technical Customer Documentation (TCD) • **See Section 3.1.2.2.1 "Power Supply"**	Assumption of Use	FUSA	ASIL B	IF
FuSa_HS_RS_449	4.3.5.0-2	Power supply of Camera Head shall not be directly connected to the vehicle 12V battery voltage. Rational: CAMERA HEAD d is NOT designed to be supplied directly from the vehicle battery and has not measures against disturbances originated from the power network of the vehicle. E.g. battery dynamic behaviour, electric pulses, reverse polarity etc.	Requirement	FUSA	ASIL B	IF

ID	Object Number	Description	Object Type	Safety Relevance	Safety Integrity	Fusa ReqType
FuSa_HS_RS_570	5	**5 Requirements - Operating the CAMERA HEAD**	Heading	No	-	-
FuSa_HS_RS_363	5.1	**5.1 Configuration of Hardware Elements**	Heading	No	-	-
FuSa_HS_RS_438	5.1.0-1	CAMERA HEAD is designed to be configured by HOST SYSTEM. Hardware components to be configured: • Imager: via I2C messages passed through the high-speed serial link • Serializer: via I2C messages passed through the high-speed serial link	Description	No	-	-
FuSa_HS_RS_367	5.1.1	**5.1.1 Imager**	Heading	No	-	-
FuSa_HS_RS_384	5.1.1.1	**5.1.1.1 Configuration of Imager**	Heading	No	-	-
FuSa_HS_RS_493	5.1.1.1-1	Info: there are two groups of Imager registers: • STATIC Registers: used for configuring the Imager operation mode and these are not changed during video stream. • DYNAMIC register: used for fine-tuning the image exposure (e.g.: analog gain, digital gain, exposure time etc…) Typically these are dynamically changed during during video streaming.	Description	No	-	-
FuSa_HS_RS_499	5.1.1.1-2	Validity of this section: STATIC register only.	Requirement	FUSA	ASIL B	Other
FuSa_HS_RS_500	5.1.1.1-3	WHEN: start-up, restarts or operation mode change during runtime (e.g.: change from 60 fps to 30 fps)	Requirement	FUSA	ASIL B	IF
FuSa_HS_RS_364	5.1.1.1-4	HOST SYSTEM shall send correct configuration data to Imager chip via I2C messages transferred via high speed serial link.	Assumption of Use	FUSA	ASIL B	IF
FuSa_HS_RS_366	5.1.1.1-5	The HOST SYSTEM shall protect the configuration data to Imager by insertion of CRC Checksum of I2C messages. note: it is safety mechanism to protect the configuration data during the transmission from HOST SYSTEM to CAMERA HEAD, finally to Imager chip.	Requirement	FUSA	ASIL B	SM
FuSa_HS_RS_472	5.1.1.1-6	OPTIONAL REQUIREMENT for ASIL B: The HOST SYSTEM can protect the configuration data sent to Imager by insertion of "Message Counter of I2C messages" in addition to the CRC Checksum. background: the message counter provides better diagnostic coverage in parallel with CRC Checksum, but not mandatory. The CRC Checksum is calculated over a known number of I2C messages, therefore a potential message loss/insertion will be detected by CRC too.	Requirement	FUSA	ASIL B	SM
FuSa_HS_RS_474	5.1.1.1-7	Details of the CRC Checksum and Message Counter: Document: [1] (Imager Safety Manual) Section: **6.1.11 [SM9, SM22] I2C CRC, I2C MESSAGE COUNTER** Page: starts on page 60	Description	No	-	-
FuSa_HS_RS_473	5.1.1.1-8	This safety mechanism can be applied in two ways, CAMERA HEAD has no preference, HOST SYSTEM can decide which option will be used. 1) INTERNAL (called "Internal" in the Imager Safety Manual) • In this case the Imager chip performs the comparison of "CRC Checksum" and "Message Counter" • If mismatch occurs in the CRC Checksum the CAMERA HEAD ensure the safe state 2) EXTERNAL (called "External" in the Imager Safety Manual) • Imager shares the CRC Checksum and the ECU performs the comparison • If CRC mismatch occurs the HOST SYSTEM shall ensure the safe state.	Requirement	FUSA	ASIL B	Other
FuSa_HS_RS_368	5.1.2	**5.1.2 Serializer**	Heading	No	-	-
FuSa_HS_RS_469	5.1.2.0-1	The Serializer can operate in two modes: "**Pixel Mode**" and "**Tunneling Mode**". Both operation mode is supported by the CAMERA HEAD and fulfil ASIL-B integrity on CAMERA HEAD level. Constrain: default mode after start-up is "**Pixel mode**". If "Tunneling mode" required, the Serializer need to be reconfigured after start-up. The HOST SYSTEM can decide which mode is used, no preference from CAMERA HEAD.	Description	No	-	-
FuSa_HS_RS_386	5.1.2.1	**5.1.2.1 Configuration of Serializer**	Heading	No	-	-
FuSa_HS_RS_332	5.1.2.1-1	HOST SYSTEM shall configure the Serializer via high-speed serial link (GMSL2). It includes: • the correct values are sent for the configuration registers • correct sequences of the messages • and other necessary requirement described in the Serializer data sheet.	Assumption of Use	FUSA	ASIL B	IF

ID	Object Number	Description	Object Type	Safety Relevance	Safety Integrity	Fusa ReqType
FuSa_HS_RS_582	5.1.2.1-2	HOST SYSTEM shall enable the internal Safety Mechanisms of the Serializer during the configuration. The exact list of Safety Mechanisms is shared separate document.	Requirement	FUSA	ASIL B	SM
FuSa_HS_RS_505	5.1.2.1-3	WHEN: start-up, restarts or operation mode change during runtime (e.g.: change from 60 fps to 30 fps)	Requirement	FUSA	ASIL B	IF
FuSa_HS_RS_371	5.1.2.1-4	HOST SYSTEM shall provide a "CRC checksum" and "Message Counter" for the configuration to ensure the integrity of messages during transmission. Safety Mechanism in the Document **[7]** • SM21 CRC on I2C and UART Transactions • SM22 Message Counter	Requirement	FUSA	ASIL B	SM
FuSa_HS_RS_528	5.1.2.1-5	The HOST SYSTEM shall check the correct operation mode (Pixel Mode or Tunneling Mode) was initiated by the Seralizer chip during start-up. Task: read out the relevant register. Value: • "Pixel Mode": expected value, normal operation. Continue the start-uo. • "Tunneling mode": abnormal condition, HOST SYSTEM shall ensure safe state.	Requirement	FUSA	ASIL B	SM
FuSa_HS_RS_529	5.1.2.1-6	The HOST SYSTEM shall read back the value of the MFP0 (IMG_RST_N) pin after the Serializer chip is intiated but before it is configured as output (push-pull) to drive the IMG_RST_N signal. Values: • "LOW" = expected value, normal operation, continue the start-up sequences. • "HIGH" = abnormal condition --> HOST SYSTEM shall ensure safe state.	Requirement	FUSA	ASIL B	SM
FuSa_HS_RS_530	5.1.2.1-7	Following ADC settings shall be set for Serializer: • Reference: internal bandgap • Input buffer: switched OFF • Input division: OFF	Requirement	FUSA	ASIL B	IF
FuSa_HS_RS_572	5.1.2.1-8	"Remote Error" functionality shall be enabled to make possible the error signalling from Serializer to Deserializer. (Enabler for FuSa_HS_RS_372)	Requirement	FUSA	ASIL B	SM
FuSa_HS_RS_448	5.1.3	**5.1.3 PMIC**	Heading	No	-	-
FuSa_HS_RS_461	5.1.3.0-1	The PMIC does not require configuration from HOST SYSTEM.	Description	No	-	-
FuSa_HS_RS_532	5.1.3.0-2	The I2C communication with PMIC is protected with CRC Checksum. Checksum calculation and checks should be activated in the HOST SYSTEM for proper operation (otherwise PMIC stops its operation). But this CRC protection is not safety related as not used in the Functional Safety of CAMERA HEAD. --> not required to implement in ASIL-B complaint	Requirement	No	-	-
FuSa_HS_RS_459	5.1.4	**5.1.4 EEPROM**	Heading	No	-	-
FuSa_HS_RS_460	5.1.4.0-1	The EEPROM does not require configuration from HOST SYSTEM.	Description	No	-	-
FuSa_HS_RS_494	5.2	**5.2 Plausibility Check of Temperature Sensors**	Heading	No	-	-
FuSa_HS_RS_506	5.2.0-1	WHEN: once in each power cycle before the video stream is used for safety related purposes (start-up, restarts)	Requirement	FUSA	ASIL B	SM
FuSa_HS_RS_495	5.2.0-2	HOST SYSTEM shall read the temperature value from the Imager and Serializer to get the temperature values.	Requirement	FUSA	ASIL B	SM
FuSa_HS_RS_496	5.2.0-3	HOST SYSTEM shall calculate the difference between the "Imager temperature" and "Serializer temperature" and then compare result against the defined thresholds. **Calculation:** $T_{delta} = T_{imager} - T_{serializer}$ where, • T_{imager}: the temperature read out from the Imager chip. (Unit: °C) • $T_{serializer}$: the temperature read out from the Serializer chip. (Unit: °C) • T_{delta}: temperature difference (Unit: °C) **Thresholds for T_{delta}:** • T_delta_LowThr: -30 (Unit: °C) • T_delta_HighThr: +30 (Unit: °C) **Reaction:** • T_{delta} is within the thresholds: normal condition, continue the start-up process • T_{delta} exceeds either the lower (T_delta_LowThr) or higher (T_delta_HighThr) threshold: abnormal condition, halt the start-up process, the HOST SYSTEM is responsible to ensure safe state.	Requirement	FUSA	ASIL B	SM

ID	Object Number	Description	Object Type	Safety Relevance	Safety Integrity	Fusa ReqType
FuSa_HS_RS_440	5.3	**5.3 Activities During Video Streaming**	Heading	No	-	-
FuSa_HS_RS_455	5.3.0-1	The items (either IF or SM) described in this section shall be activated before video streaming (more precisely when the video stream data is used for safety purposes) and shall remain active during whole streaming phase.	Description	No	-	-
FuSa_HS_RS_323	5.3.1	**5.3.1 CRC Protection for Image Data**	Heading	No	-	-
FuSa_HS_RS_324	5.3.1.0-1	The HOST SYSTEM shall check the integrity of the Image data by the CRC Checksum of the MIPI CSI2 Packets sent by the Camera Head. CRC Checksum defined in the "Low Level Layer" of MIPI CSI2 Protocol. If the two CRC checksums are not equal, reject the image frame from further processing and/or activate the safe state.	Requirement	FUSA	ASIL B	SM
FuSa_HS_RS_511	5.3.1.0-2	WHEN: continuously during video streaming when it is used safety related purposes	Requirement	FUSA	ASIL B	SM
FuSa_HS_RS_454	5.3.1.0-3	Further information in the following Section of Imager Safety Manual: Document: **[1]** (Imager Safety Manual) Section: **6.1.10 [SM8] Output Interface Fault** Page: starts on page 59	Description	No	-	-
FuSa_HS_RS_347	5.3.1.0-4	If the Error Handling Algorithm detects a faulty operation the HOST SYSTEM shall ensure the safe state.	Requirement	FUSA	ASIL B	SM
FuSa_HS_RS_350	5.3.2	**5.3.2 Frame Counter Monitoring**	Heading	No	-	-
FuSa_HS_RS_369	5.3.2.0-1	The HOST SYSTEM shall monitor the "*Frame Counter*" provided in the embedded data against the expected value. The value of counter is increased with +1 in each frame.	Requirement	FUSA	ASIL B	SM
FuSa_HS_RS_512	5.3.2.0-2	WHEN: continuously during video streaming when it is used safety related purposes	Requirement	FUSA	ASIL B	SM
FuSa_HS_RS_431	5.3.2.0-3	Details of the Frame Counter provided by the Imager: Document: **[1]** (Imager Safety Manual) Section: **6.1.12 [SM10] Frame Counter** Page: starts on page 65	Description	No	-	-
FuSa_HS_RS_432	5.3.2.0-4	If faults detected in the "Frame Counter" the HOST SYSTEM shall ensure the safe state.	Requirement	FUSA	ASIL B	SM
FuSa_HS_RS_328	5.3.3	**5.3.3 Frame Rate Monitoring**	Heading	No	-	-
FuSa_HS_RS_452	5.3.3.0-1	This safety mechanism is entirely based on the HOST SYSTEM, does not require information from the CAMERA HEAD.	Description	No	-	-
FuSa_HS_RS_508	5.3.3.0-2	WHEN: continuously during video streaming when it is used safety related purposes	Requirement	FUSA	ASIL B	SM
FuSa_HS_RS_329	5.3.3.0-3	The HOST SYSTEM shall monitor the the elapsed time between the received image frames.	Requirement	FUSA	ASIL B	SM
FuSa_HS_RS_330	5.3.3.0-4	If the elapsed time is shorter or longer than the defined value, signalling an ERROR: * High limit: YY ms (equivalent to frame rate: XYZ) * Low limit: YY ms (equivalent to frame rate: XYZ) Error Handling Algorithm (threshold number of delayed frames etc...) shall defined by HOST SYSTEM owner.	Requirement	FUSA	ASIL B	SM
FuSa_HS_RS_453	5.3.3.0-5	If the Error Handling Algorithm of "Frame Rate Monitoring" detects an faulty operation the HOST SYSTEM shall ensure the safe state.	Requirement	FUSA	ASIL B	SM
FuSa_HS_RS_488	5.3.3.0-6	note: this Safety Mechanism shall be prepared for each frame rate option: 30 Hz and 60 Hz.	Requirement	FUSA	ASIL B	SM
FuSa_HS_RS_439	5.3.4	**5.3.4 Imager - [SM7-1] SFR monitor - Imager Status Check**	Heading	No	-	-
FuSa_HS_RS_571	5.3.4.0-1	HOST SYSTEM shall monitor the status of Imager by check the status data sent in the embedded data of Image Frame. The list of the registers to be checked inside the embedded data is shared in the separate document.	Requirement	FUSA	ASIL B	SM
FuSa_HS_RS_509	5.3.4.0-2	WHEN: continuously during video streaming when it is used safety related purposes	Requirement	FUSA	ASIL B	SM
FuSa_HS_RS_451	5.3.4.0-3	Description of Safety Mechanism: Document: **[1]** (Imager Safety Manual) Section: **6.1.8 [SM7-1] SFR monitor** Page: starts on page 53	Description	No	-	-

ID	Object Number	Description	Object Type	Safety Relevance	Safety Integrity	Fusa ReqType
FuSa_HS_RS_383	5.3.4.0-4	The HOST SYSTEM shall check the CRC Checksum of the embedded data of Imager. To be clarify: is the CRC Checksum need to be checked.	Requirement	FUSA	ASIL B	SM
FuSa_HS_RS_382	5.3.4.0-5	HOST SYSTEM can check the exposure settings in the embedded data of Imager for each frame	Requirement	FUSA	ASIL B	SM
FuSa_HS_RS_478	5.3.5	**5.3.5 Imager - [SM18-2] Test Pattern (ext)**	Heading	No	-	-
FuSa_HS_RS_479	5.3.5.0-1	The HOST SYSTEM shall check the test pattern received in the embedded data of video stream.	Requirement	FUSA	ASIL B	SM
FuSa_HS_RS_580	5.3.5.0-2	VT CH3 settings: there are 4 lines of test pattern. Minimum requirement: check 2 lines (4 lines if feasible)	Requirement	FUSA	ASIL B	SM
FuSa_HS_RS_486	5.3.5.0-3	WHEN: continuously during video streaming when it is used safety related purposes	Requirement	FUSA	ASIL B	SM
FuSa_HS_RS_480	5.3.5.0-4	Description of the test pattern: Document: **[1]** (Imager Safety Manual) Section: **6.1.17 [SM18-2] TEST PATTERN (ext.)** Page: 79	Description	No	-	-
FuSa_HS_RS_475	5.3.6	**5.3.6 Imager - [SM30] HFPN DETECTOR**	Heading	No	-	-
FuSa_HS_RS_581	5.3.6.0-1	If the failure mode HFPN (Horizontal Fixed Pattern Noise) does not violate the safety requirerments in the HOST SYSTEM, the implementation of SM30 can be omitted. HOST SYSTEM owner is responsible to decide whether the implementation is required.	Assumption of Use	FUSA	ASIL B (use-case dependent)	Other
FuSa_HS_RS_476	5.3.6.0-2	The HOST SYSTEM shall implement the "HFPN Detector" algorithm to detect HFPN failures in the Image Frame.	Requirement	FUSA	ASIL B (use-case dependent)	SM
FuSa_HS_RS_485	5.3.6.0-3	WHEN: continuously during video streaming when it is used safety related purposes	Requirement	FUSA	ASIL B (use-case dependent)	SM
FuSa_HS_RS_521	5.3.6.0-4	Details of "HFPN Detector" algorithm is described in Document **[3]** (*Algorithm for HFPN Detection*). An example code in Phyton is available.	Description	No	-	-
FuSa_HS_RS_477	5.3.6.0-5	Description of the original safety mechanism: Document: [1] (Imager Safety Manual) Section: **6.1.26 [SM30] HFPN DETECTOR** Page: 101	Description	No	-	-
FuSa_HS_RS_463	5.3.7	**5.3.7 Serializer - Remote Error Detection by Deserializer**	Heading	No	-	-
FuSa_HS_RS_372	5.3.7.0-1	HOST SYSTEM shall check the "Remote Error Status" of Serializer via the Deserializer with with help of "Remote Error" function. info: CAMERA HEAD has no requirement which method (ERRB pin or status register read) is used for the Deserializer inside the HOST SYSTEM	Requirement	FUSA	ASIL B	SM
FuSa_HS_RS_507	5.3.7.0-2	WHEN: continuously during video streaming when it is used safety related purposes	Requirement	FUSA	ASIL B	SM
FuSa_HS_RS_487	5.3.7.0-3	If error is detected the HOST SYSTEM shall ensure the achievement of safe state.	Requirement	FUSA	ASIL B	SM

ID	Object Number	Description	Object Type	Safety Relevance	Safety Integrity	Fusa ReqType
FuSa_HS_RS_427	6	**6 Referenced Documents**	Heading	No	-	-
FuSa_HS_RS_428	6.0-1	**ID: [1]** Description: **Safety Manual of Imager** Title: Automotive 1/2.4" 3MP CMOS Image Sensor with On-Chip HDR & LFM Format: PDF, 141 pages Revision: v1.0.1.0 Revision Date: May 30, 2025	Description	No	-	-
FuSa_HS_RS_433	6.0-2	**ID: [2]** Description: **Camera Interface Specification** Author: Bosch Title: Camera Interface Specification Format: PDF, xx pages Revision: Revision Date:	Description	No	-	-
FuSa_HS_RS_522	6.0-3	**ID: [3]** Description: **Algorithm for HFPN Detection** Author: Samsung Title: Detection of HFPN Potentials on S5K3B6S Format: PDF, 24 pages Revision: v1.0.1.0 Revision Date: May 30, 2025	Description	No	-	-
FuSa_HS_RS_560	6.0-4	**ID: [4]** Description: **Data sheet of Imager** Author: Samsung Title: Automotive 1/2.4" 3MP CMOS Image Sensor with On-Chip HDR & LFM - Data sheet Format: PDF, 67 pages Revision: v0.08 Revision Date: July 2025	Description	No	-	-
FuSa_HS_RS_559	6.0-5	**ID: [5]** Description: **Data sheet of Serializer** Author: Analog Devices Title: MAX96717F - CSI-2 to GMSL2 Serializer Document ID: 19-100972 Format: PDF, 237 pages Revision: Rev 7 Revision Date: 4/2025	Description	No	-	-
FuSa_HS_RS_561	6.0-6	**ID: [6]** Description: **User Guide of the Serializer** Author: Analog Devices Title: MAX96717/F/R User Guide Document ID: not available Format: PDF, 116 pages Revision: version 1 Revision Date: March 7th, 2023	Description	No	-	-
FuSa_HS_RS_562	6.0-7	**ID: [7]** Description: **Safety Manual of Serializer** Author: Analog Devices Title: Safety Manual for MAX96717/MAX96717K/MAX96717F/MAX96793 (HS97) Document ID: 32-100391 Format: PDF, 52 pages Revision: Rev F Revision Date: 02/18/2025	Description	No	-	-
FuSa_HS_RS_568	6.0-8	**ID: [8]** Description: **Technical Customer Documentation (TCD)** Author: Bosch Camera Head Team Title: TCD Technical Customer Documentation Format: PDF, 49 pages Revision: 2.0 Revision Date: 2026-05-05	Description	No	-	-

ID	Object Number	Description	Object Type	Safety Relevance	Safety Integrity	Fusa ReqType
FuSa_HS_RS_456	7	**7 Document History**	Heading	No	-	-
FuSa_HS_RS_554	7.0-1	**DOORS Baseline: 18.0** Date: 2026-07-02 (yyyy-mm-dd) Author: Tibor Rácz (XC-AC/ENG2-HU) Status (draft/reviewed/released): VALID Description/Changes: • Implement the outcomes from the "4parties Meeting" safety meeting series among Bosch Camera Head Team, Bosch Image PipeLine Team, Volvo Trucks, Mekra Lang • Introduce the "Assumption of Use". It is used to describe an expected condition or behaviour from HOST SYSTEM, without detailing requirements for the vehicle architecture, implementation details etc. • Re-organizing the document structure for more logical document **Changed:** • "Intrinsic Calibration Data": it is confirmed will be used and safety relevant. Details of how is it use in the HOST SYSTEM cannot be detailed in this document, a generalized "Assumption of Use" will be use. • "FRAME_SYNC": it is confirmed: will be used in the project and NOT safety relevant • "Unique Identification Number": the details how is it used in HOST SYSTEM under clarification. Agreement: this document just offer the use, then depends on HOST SYSTEM whether apply or not. • Imager: Imager - [SM30] HFPN DETECTOR - may not need for safety (4parties meeting) • Imager: SM22 - I2C Message Counter --> became only optional requirement • "Temperature Plausibility Check" - safety relevant, shall be implemented. • Serializer: Both Pixel mode and Tunneling mode can be used from Functional Safety viewpoint. • Serializer: Insert details about configuration • Naming convention of the object ID changed from "FuSa_ECU_RS" to "FuSa_HS_RS" **Removed:** • "Camera Configuration Number" --> Removed. Reason: obsoleted, was no need for this. • Chapters "Environment Conditions" and "Product Life Cycle (Production, Service, Decomissioning)" --> removed. Reason: these requirements is in the "Technical Customer Documentation" (TCD) that is shared by Bosch to the customer. Avoid duplication.	Description	No	-	-
FuSa_HS_RS_526	7.0-2	**DOORS Baseline: 17.3** Date: 2026-06-19 (yyyy-mm-dd) Author: Tibor Rácz (XC-AC/ENG2-HU) Status (draft/reviewed/released): DRAFT Description/Changes: intermediate backup	Description	No	-	-
FuSa_HS_RS_520	7.0-3	**DOORS Baseline: 17.2** Date: 2025-10-22 (yyyy-mm-dd) Author: Tibor Rácz (XC-AC/ENH2-Bp) Status (draft/reviewed/released): DRAFT Description/Changes: Change wording, re-structure for more logical overview.	Description	No	-	-

Figure 1: From object 490 on page 3.

